

EXECUTIVE SUMMARY

High-performance full-stack engineer and systems architect specializing in real-time WebSockets, low-latency distributed microservices, and hardware-accelerated 3D WebGL interfaces. Proven track record in building sub-millisecond telemetry pipelines, Raft consensus queues, and zero-stutter 120 FPS web architectures.

TECHNICAL EXPERTISE

Languages & Core:	TypeScript, JavaScript (ESNext), Go (Golang), Rust, Python, SQL, C/C++
Frontend & Graphics:	React 19, Next.js 15, WebGL 2.0, Three.js, GLSL Fragment Shaders, Tailwind CSS, Canvas 2D
Systems & Infrastructure:	Node.js, WebSockets, gRPC, Protobuf, Redis Pub/Sub, Docker, PostgreSQL, Linux, Git

FEATURED ENGINEERING PROJECTS

PulseStream — Distributed Real-Time Telemetry Engine Engineered a distributed telemetry engine handling 100,000+ serialized msgs/sec with binary Protobuf socket brokers, achieving 0.8ms P99 packet latency and multi-threaded Web Worker offloading for 120 FPS canvas blitting.	Go • Redis • Protobuf • WebSockets
Hyperion — Volumetric 3D Raymarcher & GLSL Shaders Designed procedural atmospheric clouds and signed distance field (SDF) terrain rendered purely in GLSL fragment shaders at locked 120 FPS without loading external polygon meshes.	WebGL 2.0 • Three.js • GLSL • Linear Algebra
Aditi — High-Density Video & Real-Time Collaboration Architected multi-party conferencing platform with dynamic mesh video routing, sub-30ms signaling latency, and responsive room state orchestration.	React 19 • WebSockets • Next.js 15 • Docker
Aura — Fault-Tolerant Distributed Task Queue Implemented Raft consensus algorithm for cluster leader election, automated worker heartbeat recovery, and guarantee of exactly-once execution with 99.999% SLA.	Rust • Go • Raft Consensus • gRPC
Vector Calligraphy & Identity Engine Developed procedural vector stroke synthesizer animating mathematical cubic bezier curves with pen-nib physics, multi-ink shaders, and cryptographic verification seal generation.	Canvas 2D • Cubic Bezier Math • SVG Geometry

EDUCATION & MILESTONES

B.Tech in Computer Science & Engineering (Core) • Rungta International Skills University	2026 – 2030
Secondary School Certificate (Class 10th) • CBSE Board (Academic Excellence)	2024 • 92.0%